

1 Explainability in Reinforcement Learning

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Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the latent space learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the latent representations produced by the neural network and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representations (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the concept structure of the latent space. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

The remainder of this chapter is organized as follows. The first section introduces the theoretical assumptions underlying the proposed CAR method, referred to as the representation convergence hypothesis. The second section describes the CAR method itself, with a detailed account of the extraction of latent representations using surrogate policies, along with the automatic detection of concept structures via clustering. The third section then presents the proposed evaluation metric.

1.1 Representation Convergence Hypothesis

The Clustering Agent Representations (CAR) method relies on a assumption, referred to as the Representation Convergence Hypothesis. This hypothesis follows the two assumptions established previously throughout this manuscript.

The first assumption rejects the notion that deep neural networks constitute irreducible black boxes ???. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be highly complex, they remain explainable through the analysis of the internal mechanisms.

The second assumption states that neural networks achieve generalization by learning concepts ??. During optimization, the network progressively organizes its latent space into representations that capture the information required to solve the task.

Following these hypotheses, it becomes possible to explain the Neural Network (NN)’s process by understanding the organization of its latent space. This family of approaches is commonly referred to as mechanistic interpretability.

In mechanistic interpretability, the consistency with which a concept is recovered across different models or independent extractions is referred to as its universality [Templeton et al., 2024]. A concept is said to be universal if it repeatedly emerges despite variations in model initialization, optimization, or training conditions.

The Representation Convergence Hypothesis provides a theoretical explanation for this phenomenon. We hypothesize that, for a given task and a given neural network architecture, there exists a privileged conceptual organization. This organization naturally emerges because it constitutes an efficient representation of the information required to solve the task. Under this assumption, the universality of a concept is not merely an empirical observation but rather the expected consequence of a common underlying conceptual structure shared by independently learned representations.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a more abstract level of representation. We assume that, although the numerical implementation of the networks may differ, the conceptual organization induced by their Latent Representation (LR) converges toward an equivalent conceptual structure. Consequently, the objective is not to compare neural parameters or latent activations directly, but compare the underlying conceptual structure encoded within the latent space.

The remainder of this chapter builds upon this hypothesis. This principle forms the theoretical foundation of the *Clustering Universality* metric introduced later in Subsection 1.3.2.

1.2 Method

As discussed in Chapter ??, several hypotheses have been proposed regarding the organization of latent representations in deep neural networks. Following the work of [Zahavy et al., 2017], we therefore analyze the latent space through clustering techniques, placing our approach within the broader field of Deep Clustering (Section ??).

This design choice is motivated by our rejection of the superposition hypothesis and, more generally, of any view that assumes a linearly organized concept structure. We consider this premise unnecessarily restrictive.

The CAR methodology is composed of two main stages. The first stage

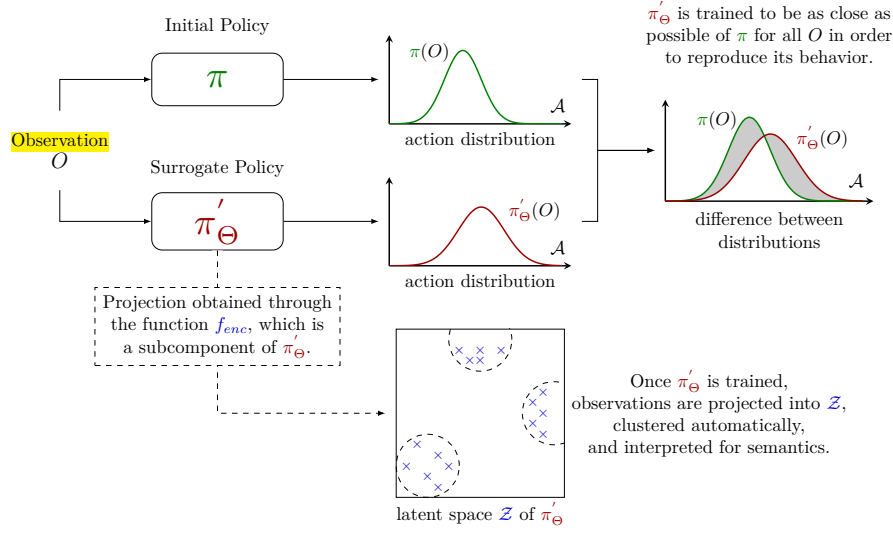


Figure 1.1: Illustration of the Clustering Agent Representation method.

consists of generating multiple independent neural networks that implement the same decision policy, referred to as surrogate policies. The second stage clusters their latent representations in order to identify the underlying conceptual organization.

This section is organized as follows. In the first subsection, we justify the use of multiple surrogate policies in order to study the stability of representations. In the second subsection, we detail how these surrogate policies are trained so as to best match the original policy.

1.2.1 Multiple surrogate policies

The CAR method was developed with the aim of verifying the representation convergence hypothesis presented in Section 1.1. To test this hypothesis, several neural networks must be trained independently and their latent representations compared. If similar conceptual structures consistently emerge, this provides empirical evidence supporting the Representation Convergence Hypothesis.

In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. Reinforcement learning, introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. Making it impossible to determine whether differences observed in the latent space originate from the learning dynamics or from genuine differences between the learned policies.

To overcome this limitation, we do not directly compare independently trained reinforcement learning agents. Instead, we first consider a initial policy that has already been trained. This policy is then imitated using several independently initialized surrogate neural networks trained through supervised behavioral cloning. Since all surrogate models approximate the same target function, they

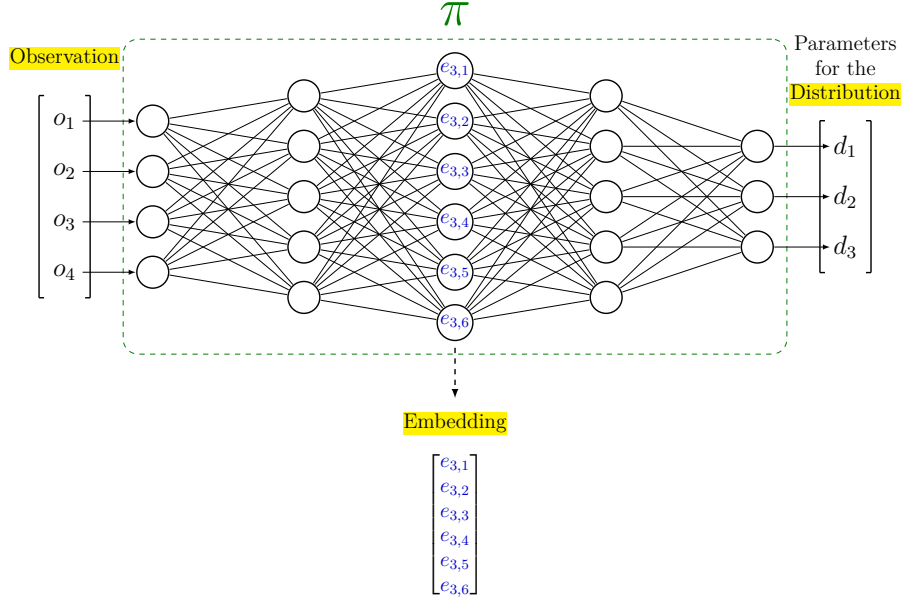


Figure 1.2: Architecture of the surrogate policy π : the encoder maps the observation O to the embedding Z , from which the decoder reconstructs the action distribution parameters.

are expected to converge toward equivalent decision policies while preserving independent internal parameterizations. It becomes possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method, as defined in Subsection ?? . The surrogate policies remain deep neural networks. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence. The explanations are therefore still extracted directly from the internal representations of deep neural networks, making CAR a fully intrinsic and mechanistic explainability approach.

1.2.2 Training surrogate policies

Throughout the remainder of this manuscript, the initial policy we aim to explain is denoted as π (represented in green in Figure 1.1). This initial policy is obtained without imposing any constraints on its origin. It may result from classical or deep reinforcement learning [?], alpha-beta algorithm [?], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π .

The replicated policies are referred to as surrogate policies, denoted by π' (shown in red in Figure 1.1). For any observation O , the distribution over the action space produced by π' should approximate that of π as closely as possible.

The surrogate policy π' is implemented as a NN parameterized by Θ . This

network is composed of two principal components: an encoder and a decoder. The encoder, represented by the function f_{enc} , maps observations from $\mathcal{O}$ into a Latent Space (LS) $\mathcal{Z}$ (shown in blue in Figure 1.1). The decoder then reconstructs actions from this latent representation. Figure 1.2 details this architecture, showing how the encoder layers progressively shape the observation into the embedding before the decoder reconstructs the action distribution parameters from it.

To train π' to accurately replicate the action distribution, we define a global loss function $\mathcal{L}$ optimized via gradient descent as introduced in Section ???. This loss, defined in Equation (1.1), is computed as the Mean Squared Error (MSE) between the parameters of the action distributions, obtained by applying the function p , which maps a distribution to its parameterization. In practice, this function does not need to be explicitly defined, as neural policies directly output the parameters of the action distribution. Once training is complete, this loss ensures that π' closely approximates the behavior of the original policy π .

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}(O) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}(p(\pi(O)), p(\pi'_{\Theta}(O))) \quad (1.1)$$

We denote by $\mathcal{E}$ the set of embeddings obtained by applying f_{enc} to a subset of observations $\mathbb{O} \subseteq \mathcal{O}$.

On the set $\mathcal{E}$, we use a clustering function f_{clust} to partition the point cloud of embedded observations into distinct clusters. Each observation is assigned a cluster label based on its proximity in the LS, resulting in the clustering $\mathcal{C}$.

To demonstrate this clustering faithfully reflects the agent’s internal decision-making process, we introduce a set of evaluation metrics, which will be detailed in the following section.

1.3 Evaluation Metrics

A key contribution of this work is the development of three evaluation metrics, designed to assess the effectiveness of CAR:

- **Behavior Reconstruction** : it measures how well a surrogate policy faithfully reproduces the behavior of the original policy by comparing their cumulative rewards.
- **Clustering Universality** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- **Abstraction Score** : it quantifies the extent to which LS clusters capture behavioral abstractions rather than directly reflecting the structure of observations or actions.

The remainder of this section details the motivation for each metric and describes how they are calculated.

The following notations will be adopted for the rest of the manuscript: we consider n surrogate policies, denoted as $\pi'_1, \dots, \pi'_n$, trained as described in Section 1.2. Each surrogate policy is associated with a projection function

173 $f_{enc_1}, \dots, f_{enc_n}$, which is applied to the set of observations $\mathbb{O}$, yielding embed-
 174 dings $\mathcal{E}_1, \dots, \mathcal{E}_n$. A clustering function is then applied to each embedding set,
 175 producing partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$.

176 1.3.1 Behavior Reconstruction

177 We argue that explaining a function b that closely approximates another func-
 178 tion a is essentially the same as explaining a . In other words, if b accurately
 179 captures the key decision patterns of a , understanding b provides meaningful
 180 insights into how a works [Guidotti et al., 2018]. In our setting, this means that
 181 generating explanations for π' that faithfully reproduce π amounts to explaining
 182 π directly. To verify that π' is indeed a faithful copy, we introduce the *Behavior*
 183 *Reconstruction* metric.

184 This metric is computed by comparing the cumulative rewards obtained by
 185 the original policy π and the surrogate policies π' over m evaluation episodes.
 186 The behavior reconstruction metric is then defined as:

$$\text{Behavior Reconstruction} = \frac{1}{n \sum_{i=1}^m g_{\pi}^i} \sum_{j=1}^n \sum_{k=1}^m g_{\pi'_j}^k \quad (1.2)$$

187 In imitation learning, it is unlikely to expect the surrogate policy π' to
 188 outperform the initial policy π it aims to approximate through supervised
 189 learning [Ross and Bagnell, 2010]. Consequently this metric ranges in $[0, 1]$.
 190 Values approaching 1 indicate minimal loss in reproducing π , while values close
 191 to 0 indicate poor reproduction.

192 We deliberately adopt a reward-based definition rather than alternatives based
 193 on actions or action distributions (e.g., training loss, accuracy, KL divergence
 194 [Shlens, 2014], or optimal transport [Peyré and Cuturi, 2020]). Since surrogate
 195 policies are trained in a supervised manner, they cannot develop strategies
 196 beyond those demonstrated by the initial policy. This means that if the rewards
 197 achieved are comparable, the surrogate policy is effectively copying the original
 198 policy’s behavior.

199 1.3.2 Clustering Universality

200 Our methodology involves analyzing the representations of an agent. Following
 201 the representation convergence hypothesis introduced in Section 1.1, for these
 202 representations to serve as a valid object of study in explainable artificial intel-
 203 ligence, they must demonstrate consistency across independent training runs.
 204 Indeed, if each surrogate π' produces substantially different representations, it
 205 is meaningless to search for recurring structures to interpret. Conversely, if
 206 independently trained surrogates yield consistent representations, the resulting
 207 structures are more likely to reflect intrinsic properties of the original policy
 208 rather than artifacts of training.

209 In our study, agent representations take the form of clusters in the latent
 210 space. To assess the stability of these representations, we measure the similarity
 211 of clusters produced by different surrogate policies π' using the Adjusted Rand
 212 Index (ARI) [Hubert and Arabie, 1985]. For this reason, we refer to our metric

as *Clustering Universality*, which evaluates the similarity between clusterings as follows:

$$\text{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^n \sum_{j=0}^{i-1} \text{ARI}(\mathcal{C}_i, \mathcal{C}_j) \quad (1.3)$$

This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate policies produce highly similar partitions, suggesting that the clusters reflect intrinsic properties of the policy. Conversely, a value near 0 implies that the similarity between partitions is no better than that of two randomly generated clusterings, meaning that the clustering fails to capture any meaningful structure in the policy. A negative value, on the other hand, signals that the agreement between partitions is worse than what would be expected by chance, indicating a systematic disagreement between the clusterings.

For further details on the properties of the ARI, we refer the reader to [Hubert and Arabie, 1985, Warrens and van der Hoef, 2022] and to Appendix ?? . Nevertheless, it is generally accepted that an ARI score above 0.5 indicates a reasonably strong agreement between two clusterings [Amezquita et al., 2020].

1.3.3 Abstraction Score

If *Clustering Universality* is high, it becomes important to determine what underlies this stability. Two possibilities arise:

- (i) the LS merely reproduces the structure of the observation space or action distribution, offering no abstraction, as defined in Definition ??;
- (ii) the LS captures similarities in decision-making processes, yielding behaviorally meaningful clusters.

To distinguish among these cases, we introduce the *Abstraction Score*. This metric contextualizes latent clusterings by examining how their patterns align with reference structures derived from the raw observation and action data. We consider : $\mathcal{C}_{obs}$, the clustering in the observation space $\mathbb{O}$, and $\mathcal{C}_{act}$, the clustering in the action-space distribution parameters. The clusterings $\mathcal{C}_{obs}$ and $\mathcal{C}_{act}$ are performed using the same algorithm, distance metric, and hyperparameters as those used for $\mathcal{C}_i$ in the LS, ensuring that any biases introduced by the clustering procedure itself are minimized.

For each latent clustering $\mathcal{C}_i$, we define its similarity to these baselines using the ARI:

$$\begin{aligned} A_i^{obs} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{obs}) \\ A_i^{act} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{act}) \end{aligned}$$

The *Abstraction Score* is subsequently calculated as one minus the average of the maximum absolute similarity values across these two reference baselines:

$$\text{Abstraction Score} = 1 - \frac{1}{n} \sum_{i=1}^n \max(|A_i^{obs}|, |A_i^{act}|) \quad (1.4)$$

246 The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the
247 latent clusters are strongly aligned with either the observation or the action
248 space, reflecting minimal abstraction, cases (i). A score close to 1 indicates that
249 the latent clusters are independent of both the observation and action spaces,
250 capturing abstractions that meaningfully reflect the agent’s behavior, case (ii).

251 The following chapter presents the results of applying these metrics to specific
252 examples.

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